

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from the specified volume and load it into Delta tables for customer and order data.

Preconditions:

- Customer data CSV file exists at `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`
- Order data CSV file exists at `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

Main Flow / Functional Steps:

- Read customer data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` CSV file.
- Load customer data into a Delta table with schema: `CustId`, `Name`, `EmailId`, `Region`.
- Read order data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` CSV file.
- Load order data into a Delta table with schema: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`.

FR-DLT-002: Transform Order Data by Adding TotalAmount Column

Title: Transform Order Data by Adding TotalAmount Column

Description: The system must add a new column "TotalAmount" to the order Delta table, which is the product of `PricePerUnit` and `Qty` columns.

Preconditions:

- Order Delta table exists with schema: `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId`

Main Flow / Functional Steps:

- Add a new column "TotalAmount" to the order Delta table.
- Calculate the value of "TotalAmount" as the product of `PricePerUnit` and `Qty` columns.

FR-DLT-003: Cleanse Customer and Order Data

Title: Cleanse Customer and Order Data

Description: The system must remove null and duplicate records from both customer and order Delta tables.

Preconditions:

- Customer and order Delta tables exist

Main Flow / Functional Steps:

- Remove null records from customer Delta table.
- Remove duplicate records from customer Delta table.
- Remove null records from order Delta table.
- Remove duplicate records from order Delta table.

FR-DLT-004: Create ordersummary Table and Load Joined Data

Title: Create ordersummary Table and Load Joined Data

Description: The system must create an "ordersummary" table if it does not exist and load joined data from customer and order Delta tables using `CustId` as the join key.

Preconditions:

- Customer and order Delta tables exist and are cleansed
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

- Create "ordersummary" table if it does not exist with schema: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.
- Join customer and order Delta tables using `CustId` as the join key.
- Load joined data into "ordersummary" table as an SCD Type 2 table.

FR-DLT-005: Update ordersummary Table on Customer Data Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the "ordersummary" SCD Type 2 table whenever there is a change in the customer table.

Preconditions:

- "ordersummary" table exists and is loaded with data
- Customer data has changed

Main Flow / Functional Steps:

- Identify changed customer records.
- Update "ordersummary" table by making old records inactive and new records active.
- Update `StartDate` and `EndDate` columns accordingly to maintain history.

FR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

Title: Create customeraggregatespend Table and Load Aggregated Data

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from "ordersummary" table.

Preconditions:

- "ordersummary" table exists and is loaded with data
- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

- Create "customeraggregatespend" table if it does not exist with schema: `Name`, `TotalAmount`, `Date`.
- Aggregate `TotalAmount` column from "ordersummary" table by `Name` and `Date` columns.
- Load aggregated data into "customeraggregatespend" table.

FR-DLT-007: Implement Data Pipeline using Delta Live Tables

Title: Implement Data Pipeline using Delta Live Tables

Description: The system must implement the data pipeline using Delta Live Tables (DLT) to perform steps 1 to 6.

Preconditions:

- All previous steps have been implemented correctly

Main Flow / Functional Steps:

- Create a DLT pipeline to perform data ingestion, transformation, and loading into Delta tables.
- Use DLT to implement SCD Type 2 logic for "ordersummary" table.
- Use DLT to aggregate data and load into "customeraggregatespend" table.